Tu Nguyen

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Education

University of Colorado Boulder — May 2028
M.S. Computer Science — GPA: 4.0/4.0

Missouri University of Science & Technology — May 2025
B.S. Computer Science — **GPA: 4.0/4.0**

Skills

Languages: JavaScript, Python, C++, C#, Java, SQL, VB | **Frameworks**: Vue.js, React, .NET, Spring Boot
Tools: Git, Docker, AutoIt, AWS, Linux, Jira | **Methods:** Agile/Scrum, REST APIs, OOP, CI/CD

Experience

Ameren Transmission — St. Louis, MO
Digital Application Development Co-op — May 2023 – December 2023

- Built UI components in **C#/.NET** for **Switching Manager**, an electrical grid management application supporting power distribution workflows.
- Wrote and optimized **SQL queries** for **Switching Manager**, reducing query response time by **40%** and improving data retrieval for grid-management workflows.
- Collaborated with senior developers to implement and test new features under established coding standards in a production software environment.
- Automated repetitive **Excel-based data processing tasks** using **AutoIt**, reducing manual processing time by two weeks for engineering or operations staff.

Projects

AI Note-Taking App — Personal Project (In Progress) — January 2026 – Present

- Building a Vue.js AI app that converts transcript text, uploaded files, and audio/video into structured Markdown notes for Notion and Obsidian.
- Designing transcription and note-generation REST APIs with a consistent Markdown schema, regenerate-from-source workflows, and split-screen review/editing.
- Targeting 95% schema consistency and sub-30s/sub-90s generation time for transcript and media inputs across all 3 MVP input paths.

Missouri University of Science and Technology — Rolla, MO
Software Engineer — Boeing-Sponsored Capstone Project — January 2025 – May 2025

- Developed a **C++ path-finding system** for a Boeing-sponsored air-to-ground search project to cover **80%** of a **2D grid world** while obeying **aircraft fuel, obstacle, and movement constraints**.
- Implemented route optimization to **minimize cell movements** and improve search efficiency in a constrained grid-based simulation.
- Built **dead-area detection** using a **water-drop algorithm** to identify obstacle-surrounded spawn regions and prevent invalid aircraft placement.
- Iterated on routing logic across **Scrum** milestones to improve path selection and simulation usability.

Missouri University of Science and Technology — Rolla, MO
Team Lead — Software Engineering Capstone — August 2024 – December 2024

- Led a **3-person team** in building a **React-based spaced-repetition PDF viewer** that supported annotation, summarization, and interactive document review.
- Engineered React components to support rendering and annotation for **500+ page PDFs**.
- Architected a scalable frontend workflow with code reviews and 85% test coverage.

Certifications

- Gen AI for Data Engineers Spec (Feb 2026)
- Google AI Essentials (Feb 2026)
- Vue: The Complete Guide (Mar 2026)
- **AWS Certified Developer – Associate (Mar 2026)**